Yeast Fermentation

Teacher guide | Balloon lab | Biology 2e, Chapter 7

1. Organize a full comparison set

Question: How does the available sugar affect gas accumulation in a yeast culture? This download is specifically for balloon lab. Use the same method for all flasks in a pooled comparison.

Equipment and measurement

Matched flasks, same-type balloons that fit the necks, flexible tape or string and ruler. Also use a timer, thermometer, labeled transfer tools, stocks and eye protection.

A balloon collects gas. Measure the distance around its widest part. Circumference is a relative indicator of gas accumulation, not a volume in milliliters.

One set = 9 independent flasks: N1-N3 (no added sugar), G1-G3 (glucose), S1-S3 (sucrose). Each flask is a fresh mixture. Multiple readings from one flask are not additional independent trials.

Students	Example allocation of the 9 flasks
18: nine pairs	Each pair prepares 1.
12: six pairs	Three pairs prepare 2; three pairs prepare 1.
10: five pairs	Four pairs prepare 2; one pair prepares 1.
9: three pairs and one trio	Each team prepares 2; the teacher prepares the ninth. Or assign the trio 3.

Assign one or two flask IDs per pair and record ownership on page 4. Roles: mix/prepare, measure/time, record/share. Pairs can combine roles. Each flask needs its own start time and elapsed-time record.

Demo: distribute observation and analysis roles while the teacher operates equipment. If using just one flask per treatment, report n = 1; do not call three time readings three trials. Share matched independent class trials to complete a set.

Allow about 10-15 minutes to plan/prepare, 30 minutes to observe, and 15-20 minutes to pool/analyze. Use a second lesson for analysis if needed. Pilot the equipment and small-flask volumes before class.

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2. Stocks, flask sizes, and supply planning

Choose one flask size and recipe per pooled set. These are proposed pilot volumes, not a promise of measurable gas in one lesson. Keep all treatments at the same measured temperature; a teacher-managed 30 C water bath is one possible condition.

Flask capacity	Treatment stock*	Yeast suspension	Final liquid
50 mL	10 mL	10 mL	20 mL
100 mL	20 mL	20 mL	40 mL
Custom: ____ mL	____ mL	____ mL	____ mL

*Plain water for N; glucose stock for G; sucrose stock for S. Examples leave 60% headspace for gas/foam. Lower the fill if the pilot requires it; do not fill a 50 mL flask with 50 mL culture.

Optional stock recipe

For each sugar: dissolve 80 g glucose OR sucrose and bring to a final volume of 1,000 mL with water (8% mass/volume). Disperse 20 g dry baker's yeast and bring to 1,000 mL (2% suspension). Keep plain water separately. Equal volumes of sugar stock and yeast give 4% sugar and 1% yeast in the flask. N has no added sugar.

These match the virtual mass concentrations, not its 50 mL culture volume. Record the glucose product (including any dextrose monohydrate). Equal sugar mass is not equal molarity. Teachers may substitute a piloted recipe.

Prepare labeled sugar stocks ahead. Prepare yeast close to use following package hydration guidance. Do not add sugar to shared yeast stock. Gently resuspend before EVERY aliquot because yeast settles. Use separate labeled transfer tools and equilibrate all stocks to the chosen test temperature.

For one 9-flask set	50 mL flasks	100 mL flasks
Plain water / glucose / sucrose (each)	30 mL	60 mL
Yeast suspension (all 9)	90 mL	180 mL
Sets from 1 L of EACH stock*	11	5

*Before losses; yeast is limiting. For B sets and V mL of each stock per flask: each treatment stock = 3 x B x V; yeast = 9 x B x V. Allow extra for spills, pilot and optional controls.

Chosen sugar stock: ____ g / ____ mL. Yeast: ____ g / ____ mL. Final sugar concentration: ____ %. Test temperature: ____ C.

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3. Set up, measure, and interpret

Check balloon fit and latex allergies. Use the same teacher-selected balloon preparation for every setup; do not inflate by mouth. Fit the balloon promptly after adding yeast.

Label assigned flasks; dispense treatment stock first. Gently resuspend yeast stock immediately before each aliquot. Add yeast last, mix consistently, promptly connect the collector and start that flask's timer. Record the initial reading and any connection delay.

Read at 0, 5, 10, 15, 20, 25 and 30 elapsed minutes, or a shared teacher-selected schedule. Gently measure the widest circumference without squeezing. Use the same measurement method each time. Record NI (not inflated) for a limp balloon; do not invent a zero.

Record actual circumference when measurable; record NI when limp. Do not subtract NI as if it were zero. NI = not inflated; M = missing; L = suspected leak. Do not replace NI or M with zero.

Pooling and interpretation

Average only comparable numeric circumferences; report n and the number NI at each time. If NI occurs, label any numeric-only mean; it does not represent all three flasks. Keep raw readings and flag leaks, missing data and any exclusions; do not silently discard unusual results.

Elasticity, inflation threshold, leaks and temperature affect the measurement. Do not convert circumference to volume using a spherical-balloon formula. A cm/min change is not a gas-production rate.

Keep recipes, temperature, flask size, apparatus and units matched. Keep virtual and physical results separate even if both use mL. For an interval rate, divide the change in circumference by elapsed minutes (cm/min). A late interval rate and total gas accumulated are different quantities.

Do not require a winning sugar. Gas accumulation supports a claim about yeast activity; it does not measure ATP yield or establish oxygen absence. The no-added-sugar control can show background activity. An optional sugar-without-yeast control helps check physical expansion.

Practical checks

Wear eye protection; check glassware and follow school procedures. Do not taste cultures. Teacher manages warm water, foaming, cleanup and equipment capacity. Never use a rigid cap. The teacher stops setups if foam enters a balloon or its inflation becomes excessive.

If readings do not change: check yeast freshness, temperature, settling, leaks and collector performance. For balloons, inflation threshold may hide small gas amounts. Keep observations, allow more time if practical, and document problems.

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4. Assignment roster and sharing

Class/date: _____ Set ID: _____ Students: ____

Method: Balloon lab | Unit: cm

Flask: ____ mL Treatment stock: ____ mL Yeast: ____ mL

Temperature: ____ C Reading interval: ____ min End time: ____ min

Flask ID	Treatment	Pair/team or demo	Start time
N1	No added sugar		
N2	No added sugar		
N3	No added sugar		
G1	Glucose		
G2	Glucose		
G3	Glucose		
S1	Sucrose		
S2	Sucrose		
S3	Sucrose		

Print worksheet pages 1-2 for each pair/team and pages 3-4 for each student or pooled group. Page 2 records up to two flasks; copy it for additional flasks. Page 3 combines all nine.

Report actual n if fewer than three valid independent trials remain. Keep flagged originals if a fresh replacement is made. Use the classroom worksheet or a separate spreadsheet. The virtual lab stays on a gas syringe; never enter balloon cm into its mL fields.

Method and reading connections

Science World: Yeast-Inflated Balloons https://www.scienceworld.ca/resource/yeast-inflated-balloons/ Science Buddies: An Aerobic Exercise (inverted-cylinder collection) https://www.sciencebuddies.org/science-fair-projects/project-ideas/MicroBio_p009/microbiology/yeast-metabolism-aerobic-anaerobic OpenStax Biology 2e, 7.5: Metabolism without Oxygen https://openstax.org/books/biology-2e/pages/7-5-metabolism-without-oxygen

The small-flask recipes and volumes are proposed pilot settings, not validated performance claims. The virtual apparatus measures collected gas; it does not reproduce every physical measurement error.